

Agentic AI for Clinical Decision Support (CDSS)

Advanced Pattern Recognition & Reasoning
for TPO Optimization

PRESENTER

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TOPIC

Topic 2 - Clinical Decision Making



Beyond Generative AI: The Agentic Shift



Clinical Decision Making

The computational evaluation of patient data to drive diagnoses and clinical interventions.



Agentic Generative AI

Goal-directed systems that retrieve data, reconcile evidence, and execute decisions autonomously.



TPO Integration

Automating reasoning behind Treatment (T), Payment (P), and Operations (O) to eliminate Black Box uncertainty.

KEY INSIGHT: Static models predict; Agentic models reason and justify.

Market Drivers & Explainability

\$4.46B

Healthcare Agentic AI
Market by 2030

44.8% CAGR

Compound Annual
Growth Rate

Source: Market Research 2025

⚠ The Hallucination Barrier

Traditional LLMs risk generating plausible but incorrect clinical outputs. Retrieval-Augmented Generation (RAG) is now the accepted clinical safety standard to ground outputs in verified evidence.

✓ Explainability (XAI) as a Mandate

SHAP and LIME are no longer optional - they are required for clinician trust and regulatory compliance. The 2025 PMC Review confirms XAI adoption as a non-negotiable standard in production AI systems.

Technical Stack: The 3-Layer Logic

01

Predictive Layer

Ensemble ML

XGBoost + Random Forest
for pattern recognition

02

Explainability Layer

SHAP Integration

Local feature transparency
for each prediction

03

Reasoning Layer

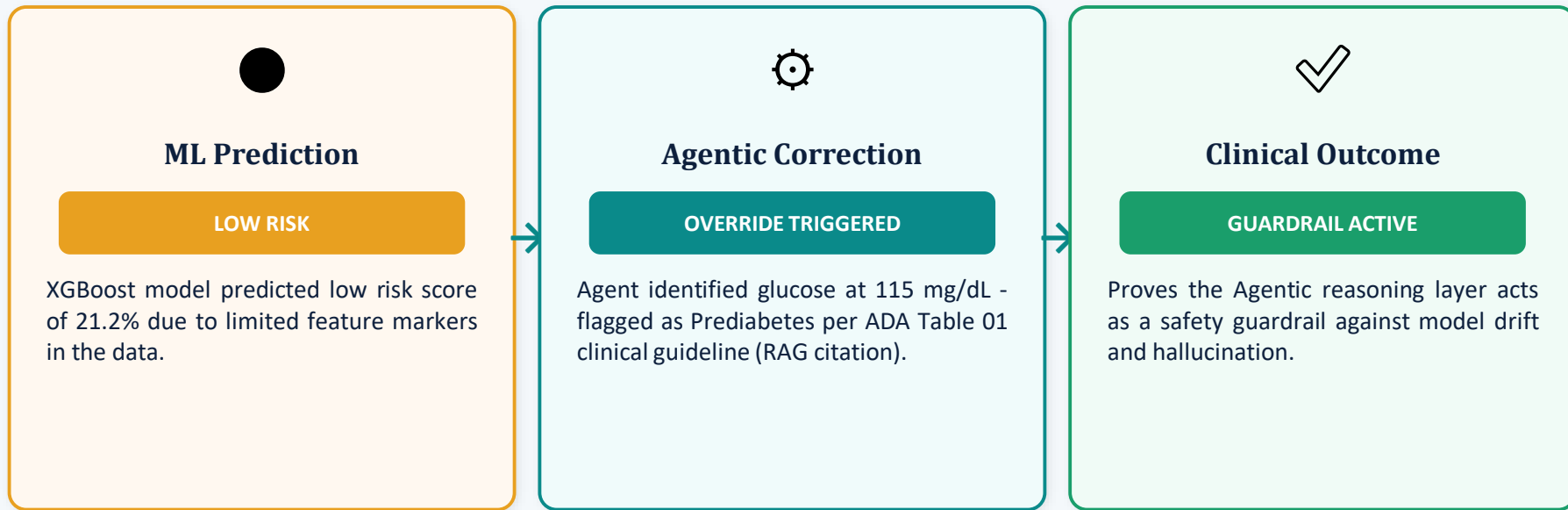
Llama-3.3-70B via Groq LPU(TM)

Grounded in ADA 2023
Clinical Guidelines (RAG)

KEY ADVANTAGE: High-speed inference (Groq LPU) combined with strict ADA 2023 medical grounding via RAG

Edge-Case Intelligence: Reconciling ML with Guidelines

CASE STUDY: Prediabetes Edge-Case Detection



CLINICAL SAFETY PROOF: The Agentic reasoning layer prevents diagnostic errors that rule-based models cannot catch.

Strategic Business Analysis

Cotiviti · Agentic AI CDSS

OPPORTUNITIES

✓ Payment Integrity

40-60% reduction in administrative costs via automated claim-guideline cross-referencing, directly improving Cotiviti's core revenue model.

✓ Scale & Standardization

Standardizing CDSS capabilities across all clinical specialties, creating a unified AI platform advantage over point solutions.

THREATS

⚠ Regulatory Ambiguity

Lack of unified clinical AI standards creates liability risk and may delay deployment timelines across regulated payer markets.

⚠ Platform Consolidation

Market dominance by Microsoft/Epic requires Cotiviti to differentiate via deep clinical domain expertise and proprietary data assets.

Future-Proofing Cotiviti's AI Roadmap

01

Deploy RAG-Grounded Reasoning

Move from rule-based engines to auditable reasoning agents for claim review. RAG-grounded outputs provide citation trails for compliance and clinician trust.

02

Establish Governance Standards

Mandate SHAP explainability and RAG citation for all production AI. Standardized governance reduces liability and accelerates regulatory approval.

03

Invest in Specialized Talent

Build a Clinical AI Research Pipeline to capture scarce engineering/medical hybrid talent before competitors can consolidate the market.